Treating diseases

9C Threat from disease · Lesson 4 of 7

Date _____ Class _____

Today I can 今天我能

- ☐ I can explain which medicines treat which types of disease, and why antibiotics do not work on viruses.
- ☐ I can explain how large amounts of human hormones are made quickly using genetically modified bacteria in a fermenter.

What happens today

1. Do Now: which treatment?
2. Learn: Drugs and medicines · Diseases we cannot cure · Transgenic bacteria · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 44 to 45 · Workbook pages 32

What I learned today (one or two sentences) 今天我学到了：

Key Vocabulary

Term	中文	Definition
drug	药物	a substance that affects how the body works
medicine	药品	a drug used to treat or prevent disease
antibiotic	抗生素	a medicine that kills bacteria
antiviral	抗病毒药	a medicine that stops viruses replicating (copying themselves)
genetic disorder	遗传病	a disease caused by faulty genes that a person is born with
autoimmune disease	自身免疫病	a disease in which white blood cells attack the body's own cells
insulin	胰岛素	a hormone made in the pancreas that lowers blood glucose
transgenic	转基因的	contains DNA from more than one species
fermenter	发酵罐	a large container in which microorganisms are grown
sterilise	灭菌	kill all the microorganisms in or on something
aseptic	无菌的	free from unwanted microorganisms

Command words 指令词: Complete 完成（填写） · Describe 描述 · Draw 画出 · Explain 解释 · Give 给出 · Match 连线（配对） · Tick 打勾

Quick Review

which treatment?

1 An ear infection is caused by bacteria. Would you give an antibiotic or an antiviral?

2 Scurvy is caused by too little vitamin C. How would you treat it?

3 A vaccine prevents a disease. Can it treat someone who already has it?

Core 1 of 2

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 Match each scientific word to its definition. Draw a line.

medicines	drugs that prevent viruses replicating
antibiotics	a hormone that makes cells take up glucose from the blood
antivirals	drugs used to treat or prevent disease
vaccines	substances that make us immune to particular diseases
insulin	drugs that kill bacteria

2 Tick one box in each row to show the best treatment for each disease.

Disease	antibiotic	antiviral	add a nutrient	inject a hormone	vaccine (prevents it)
ear infection caused by bacteria					
flu (caused by a virus)					
scurvy (too little vitamin C)					
type 1 diabetes (no insulin made)					
measles, before a child catches it					

Core 2 of 2

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 Complete the text about fermenters using words from the word bank.

Transgenic bacteria are grown in a large container called a _____. Before use it is _____ with steam so that no other _____ can grow inside it. _____ air is bubbled in so that the bacteria can respire. _____ paddles mix the bacteria, nutrients and oxygen, and a water _____ keeps the temperature steady.

Challenge 1 of 2

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 Transgenic bacteria are grown in fermenters.

a) Give the reason why air is bubbled into the fermenter.

b) Describe two ways in which the fermenter keeps the bacteria growing at a high rate.

Challenge 2 of 2

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 *continued*

c) Describe the aseptic precautions taken when preparing the fermenter, and explain why they are needed.

C2 Scientists can also remove genes from an organism. Explain how this can create a genetically modified organism that is not transgenic.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is the difference between an antibiotic and an antiviral?

2 What does 'transgenic' mean?

3 Give one reason why a fermenter is sterilised before use.

Today I can: Goal 1 ☐ Goal 2 ☐